



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year: 2025-2026 (Even Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member: Dr. S. Poornam

Date & Time: 28.01.2026 & 11:35 A.M

Topic: Rijk method

Course Outcome: CO2

Type of Activity: Engaged Learning Through Challenges

Justification

This activity was designed to enhance students' understanding of the Rijk Method in Theory of Computation through challenge-based learning. Students often find formal methods and automata-related procedures difficult due to abstract concepts and sequential problem-solving steps. By introducing learning challenges, puzzles, and group-based analytical tasks, students became actively involved in the learning process. The engaged learning approach promoted critical thinking, logical reasoning, collaboration, and confidence in solving complex computation problems.

Activity Description

Engaged Learning Through Challenges: Rijk Method

Introduction Phase

- ❖ The instructor introduced the fundamentals and applications of the Rijk Method in Theory of Computation.
- ❖ Students were briefed about the challenge-based learning activity and its objectives.
- ❖ The class was divided into small groups for collaborative participation.

Challenge Activity Phase

- ❖ Each group was given structured problems related to the Rijk Method.
- ❖ Students analyzed patterns, identified rules, and applied appropriate computation techniques.
- ❖ Challenge cards containing different complexity levels were distributed:
 - Basic Concept Challenges,
 - Intermediate Analytical Problems,
 - Advanced Application Tasks.
- ❖ Students competed to solve the problems within the given time limit.
- ❖ Peer interaction and collaborative reasoning were encouraged throughout the activity.

Discussion and Solution Presentation

- ❖ Groups presented their solutions and explained their reasoning steps.
- ❖ Students compared alternative approaches for solving the same problem.
- ❖ The instructor clarified misconceptions and highlighted efficient solution strategies.

Reflection and Assessment

- ❖ Students completed a short quiz based on the activity.
- ❖ A reflective discussion was conducted on:
 - Challenges faced during problem-solving.
 - Logical strategies used.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3
CO1	2	2	2	1	2	1	2

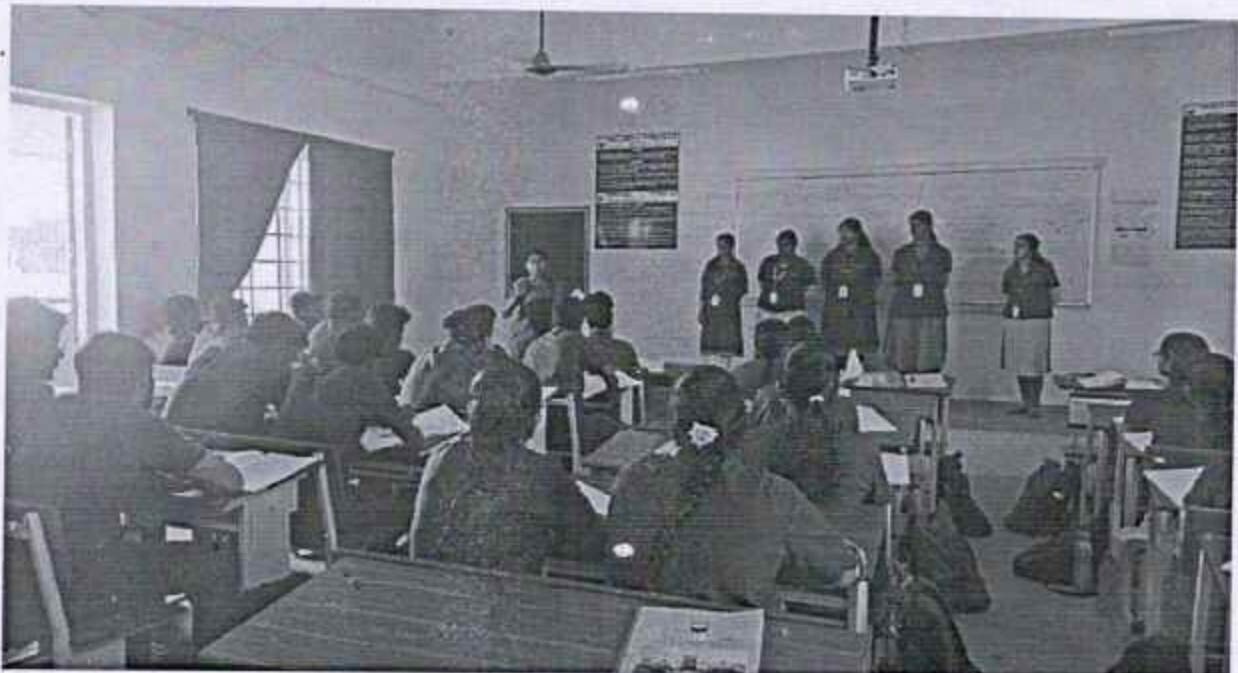
(1 – Low, 2 – Moderate, 3 – High)

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1, PSO2, PSO3

Justification for correlation

- Students applied theoretical computation concepts to solve analytical challenges.
- Challenge-based activities improved logical thinking and problem-solving skills.
- Group discussions enhanced communication and collaborative learning abilities.
- Students gained practical understanding of the Rijk Method through active participation.
- The activity strengthened analytical reasoning and computational thinking.

Images / Screenshot of the practice:





Outcome

- Students demonstrated improved understanding of the Rijk Method concepts.
- They actively participated in analytical and problem-solving activities.
- Challenge-based learning increased engagement and classroom interaction.
- Students developed confidence in solving Theory of Computation problems.
- The activity encouraged teamwork, logical reasoning, and peer learning.

Identified Challenges:

- Students found formal computation methods abstract and difficult to visualize.
- Lack of confidence in solving computation problems independently.
- Reduced classroom interaction during traditional lecture sessions.

Addressing Challenges:

- Introduced challenge-based engaged learning activities.
- Designed structured problems with varying difficulty levels.
- Encouraged collaborative discussion and peer-supported learning.

Post-Implementation

- Students showed enthusiastic participation throughout the activity.
- Group challenges improved interaction and analytical thinking.
- Most students were able to apply the Rijk Method correctly after discussion sessions.
- Peer collaboration helped students understand problem-solving steps effectively.
- Feedback indicated that challenge-based learning made Theory of Computation more interesting and understandable.

References

- <https://www.geeksforgeeks.org/theory-of-computation-automata-tutorials/>
- https://www.tutorialspoint.com/automata_theory/index.htm
- Recentering Academics | The Derek Bok Center for Teaching and Learning
- <https://www.edutopia.org/article/challenge-based-learning-real-world>

Signature of Faculty Member

HOD